

SECURE AND REVERSIBLE DATA EMBEDDING IN MEDICAL IMAGES USING U-NET SEGMENTATION AND XOR ENCRYPTION

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Abstract—The rapid growth of digital healthcare systems has created a pressing need for secure storage and transmission of medical images. Ensuring data confidentiality while preserving diagnostic image quality is a fundamental requirement for medical image communication. This paper presents a secure and reversible data embedding framework for brain tumor Magnetic Resonance Imaging (MRI) images. The proposed framework integrates U-Net-based tumor segmentation to localize the Region of Interest (ROI) for guided embedding, a Least Significant Bit (LSB)-based reversible data hiding technique to embed patient information, and XOR-based image encryption for transmission security. The reversibility of the proposed method is guaranteed by storing the original LSBs prior to embedding and reapplying them during restoration. Experiments conducted on the BraTS dataset yield a Peak Signal-to-Noise Ratio (PSNR) of 73.83 dB, a Structural Similarity Index (SSIM) of 0.99999, and a Mean Squared Error (MSE) of 0.00269, confirming near-lossless reconstruction suitable for clinical diagnosis. The proposed method demonstrates superior performance over existing reversible data hiding methods in terms of image quality metrics.

Keywords— MRI, Reversible Data Hiding (RDH), Medical Image Security, U-Net, Image Encryption, Image Restoration, Convolutional Neural Networks (CNN), LSB Embedding.

I. INTRODUCTION

MRI scans are widely used for brain tumor detection because they provide excellent soft-tissue contrast without ionizing radiation. In the era of telemedicine, cloud computing, and electronic health records, MRI images are routinely shared across networks, raising serious concerns about patient privacy and data integrity. Unauthorized access or modification of medical images may result in privacy breaches or diagnostic errors [1].

Conventional encryption methods protect image data during transmission, but they do not support embedding patient metadata within the image itself. As a result, patient records and images are often stored separately, creating a risk of misassociation and information loss. Moreover, irreversible data hiding methods introduce permanent distortions that can compromise image quality and diagnostic value [2].

Reversible Data Hiding (RDH) addresses these limitations by allowing patient data to be embedded within the host image while enabling both the original image and the embedded data to be perfectly recovered at the receiver's end [3]. Achieving

an optimal trade-off between embedding capacity, security, and imperceptibility remains, however, a significant research challenge [4].

Recent advances in deep learning (DL) have substantially improved medical image processing. The U-Net architecture, in particular, has demonstrated strong performance in medical image segmentation by combining encoder-decoder pathways with skip connections for accurate spatial feature recovery [5], [6]. Embedding patient data within the tumor ROI, as identified by U-Net, limits distortion to a clinically less-sensitive region, thereby preserving overall diagnostic quality. MRI images are also susceptible to acquisition noise; hence, preprocessing steps including denoising are essential prior to embedding [7].

The proposed method guarantees reversibility through the following mechanism: before embedding, the original LSBs of the selected tumor-region pixels are recorded and stored as side information. During restoration, these original LSBs are reappplied to the decoded image, ensuring pixel-perfect reconstruction with zero information loss. This distinguishes the approach from irreversible methods that permanently alter pixel values.

The main contributions of this paper are:

- A U-Net-guided ROI-based LSB embedding strategy that confines data insertion to the tumor region, minimizing perceptual distortion in diagnostically sensitive areas.
- A mathematically guaranteed reversibility scheme using original-LSB preservation, achieving $MSE = 0$ and $PSNR = \infty$ at the restoration stage.
- XOR-based encryption of the stego image prior to transmission, ensuring embedded patient information remains confidential.
- Comprehensive evaluation on the BraTS dataset with PSNR, SSIM, and MSE, and comparative analysis against nine state-of-the-art RDH methods.

II. LITERATURE SURVEY

A CNN-based image steganography framework is presented in [1], which uses deep networks for both embedding and extraction. The method achieves higher embedding capacity and superior stego image quality compared to the classical LSB approach, but does not address medical image reversibility.

A residual learning-based deep convolutional neural network (DnCNN) for image denoising is introduced in [2]. The DnCNN surpasses conventional filters such as BM3D in denoising performance; however, it demands large training datasets and significant computational resources.

An adaptive moment estimation optimizer (Adam) for deep network training is described in [3], improving convergence stability in image restoration networks. Hyperparameter sensitivity remains a practical limitation.

A deep residual learning framework for image representation is proposed in [4], which improves convergence and addresses gradient degradation for very deep networks. The method delivers strong results on large-scale datasets at the cost of increased model complexity.

Collaborative 3D-transform-domain filtering via the BM3D algorithm is analyzed in [5]. BM3D achieves high PSNR for Gaussian noise, but its performance degrades for complex or mixed-noise distributions.

A comprehensive review of DL-based image steganography is presented in [6], showing that DL models automatically learn optimal embedding patterns, improving resilience and reducing detectability compared to hand-crafted methods.

A deep residual network for steganalysis is proposed in [7], achieving high detection accuracy for hidden data in digital images, underscoring the importance of designing secure embedding strategies.

A novel U-Net-based CNN for image segmentation and restoration is introduced in [8], demonstrating strong segmentation performance, though the approach is susceptible to overfitting when high-quality annotated datasets are limited.

A CNN-based image super-resolution network is presented in [9] as one of the first DL-based image restoration frameworks. While effective for resolution enhancement, the network performs poorly on degraded or noisy inputs.

The survey reveals that no existing method simultaneously combines U-Net-guided ROI localization, reversible LSB embedding with original-LSB preservation, and XOR encryption in a unified framework for medical images. The proposed method addresses this gap.

III. PROPOSED METHODOLOGY

The proposed framework for secure and reversible data embedding in brain tumor MRI images maintains both data confidentiality and diagnostic image quality throughout the pipeline. The complete workflow is illustrated in Fig. 1 and consists of seven stages: (1) image preprocessing and resizing, (2) noise estimation and reduction, (3) grayscale verification, (4) U-Net-based tumor segmentation and reversible LSB embedding, (5) XOR encryption and Base64 encoding, (6) decryption and data extraction, and (7) original image restoration. Reversibility is guaranteed by storing the original pixel LSBs prior to embedding and restoring them in step (7), yielding zero reconstruction error.

A. Image Preprocessing and Resizing

An MRI volume is selected from the BraTS dataset in NIfTI format. A single axial slice from the center of the 3D volume

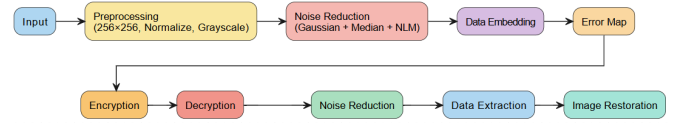


Fig. 1. Overall flowchart of the proposed secure and reversible data embedding framework. The pipeline proceeds from MRI acquisition through preprocessing, U-Net segmentation, LSB embedding, XOR encryption, transmission, decryption, data extraction, and lossless image restoration.

is extracted and converted to a 2D array. The slice is resized to 240×240 pixels for uniform processing. Pixel intensities are normalized to the range $[0, 255]$:

$$I_{\text{norm}} = \frac{I - I_{\min}}{I_{\max} - I_{\min}} \times 255$$

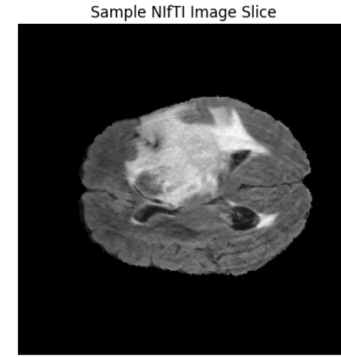


Fig. 2. Input MRI slice extracted from the BraTS NIfTI volume and resized to 240×240 pixels.

B. Noise Estimation

Noise is estimated using two complementary statistical measures. The standard deviation (SD) captures global intensity variation, while the Laplacian variance captures local high-frequency content attributable to noise and edges. The SD is:

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2}$$

The Laplacian operator is applied as a second-order spatial derivative:

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

C. Noise Reduction

Three denoising filters are evaluated: (1) *Gaussian* filtering, which suppresses noise via weighted spatial averaging; (2) *Median* filtering, which replaces each pixel with the neighborhood median, preserving edges; and (3) *Non-Local Means* (NLM) filtering, which exploits self-similarity across the image to reduce noise while retaining fine structural detail such as tumor boundaries.

Quantitative results are given in Table I. The NLM filter is selected for all subsequent experiments because it achieves

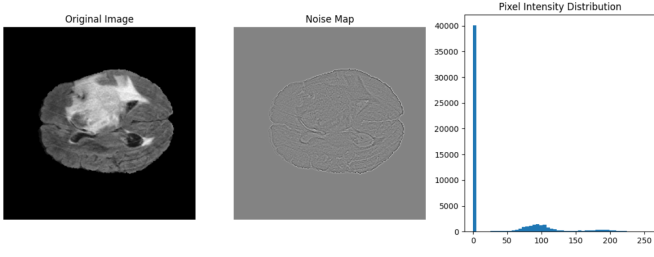


Fig. 3. Noise visualization: (left) noise map and (right) intensity histogram of the input MRI slice. SD = 58.12; Laplacian variance = 509.21.

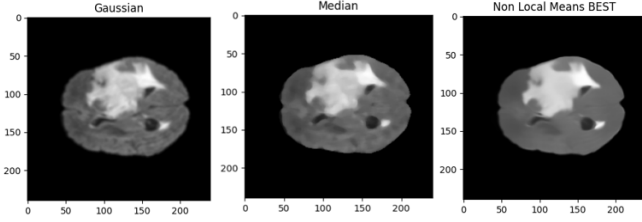


Fig. 4. Comparison of three denoising techniques applied to the input MRI image: (a) original, (b) Gaussian, (c) Median, and (d) Non-Local Means (NLM). NLM best preserves tumor boundary sharpness while reducing noise.

the best balance between noise suppression (SD = 56.99) and edge preservation (Laplacian variance = 32.64).

TABLE I
QUANTITATIVE COMPARISON OF DENOISING TECHNIQUES

Method	SD	Laplacian Variance
Original	58.12	509.21
Gaussian	56.91	9.08
Median	57.17	155.43
Non-Local Means	56.99	32.64

D. Grayscale Verification

The denoised image is verified to be single-channel by inspecting the array shape: (240,240) confirms grayscale, while (240,240,3) would indicate RGB. The verified image has shape (240,240), type `uint8`, and intensity range [0, 232].

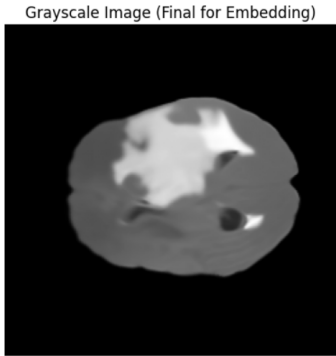


Fig. 5. Grayscale MRI image after preprocessing. Shape: (240, 240), Type: `uint8`, Range: 0–232.

E. Reversible Data Hiding via U-Net-Guided LSB Embedding

1) *U-Net Tumor Segmentation*: The U-Net encoder applies successive convolutional and max-pooling layers, with filter counts increasing from 64 to 512 across four encoder blocks. Each convolutional layer computes:

$$F(x, y) = \sum_{i=0}^{k-1} \sum_{j=0}^{k-1} I(x+i, y+j) \cdot K(i, j)$$

with ReLU activation $\text{ReLU}(x) = \max(0, x)$. Max-pooling halves the spatial resolution:

$$P(x, y) = \max(I_{\text{region}})$$

The decoder restores the feature map to 240×240 via transposed convolutions with skip connections. A final 1×1 sigmoid layer produces a probability mask; pixels with probability ≥ 0.5 are classified as tumor ROI.

2) *Reversible LSB Embedding with Original-LSB Preservation*: Before embedding, the LSB of each selected tumor-region pixel is read and stored as side information $\mathbf{b}_{\text{orig}}$, which is the key to reversibility. Patient data are converted to 8-bit ASCII binary:

$$B_i = \text{format}(\text{ASCII}(c), 8)$$

Each data bit b is embedded by replacing the pixel's LSB:

$$P'(x, y) = (P(x, y) \& 254) \mid b$$

Since only the LSB is modified, the maximum per-pixel intensity change is 1 gray level, ensuring visual imperceptibility. At restoration, the original LSBs are reapplied:

$$I_r(j, k) = (I_e(j, k) \& 254) \mid b_{\text{original}}(j, k)$$

This guarantees pixel-perfect restoration with MSE = 0.

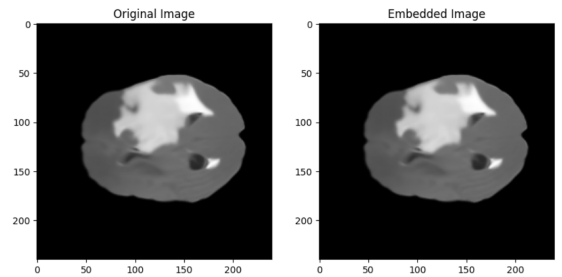


Fig. 6. Stego image after LSB embedding of patient data in the U-Net-identified tumor region.

3) *Embedding Distortion Analysis*: The pixel-level distortion introduced by embedding is:

$$D(a, b) = |I_{\text{original}}(a, b) - I_{\text{embedded}}(a, b)|$$

Since only LSBs are modified, $D(a, b) \in \{0, 1\}$. Non-zero values are confined to the tumor ROI, as confirmed by the error map in Fig. 7.

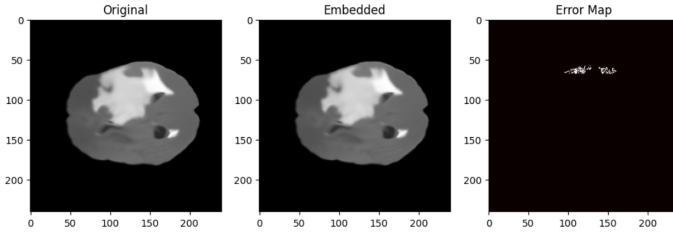


Fig. 7. Error map between the original and stego images. Non-zero values (maximum = 1) are confined to the tumor region, confirming negligible and localized embedding distortion.

F. Image Encryption and Transmission

The stego image is encrypted using bitwise XOR with a pseudo-random key stream K :

$$E(m, n) = I(m, n) \oplus K(m, n)$$

XOR encryption is computationally efficient and produces a ciphertext visually indistinguishable from random noise (Fig. 8). The encrypted image is serialized to a byte array and encoded in Base64 for text-safe storage and transmission:

$$B = \text{bytes}(E), \quad T = \text{Base64}(B), \quad S = \text{String}(T)$$

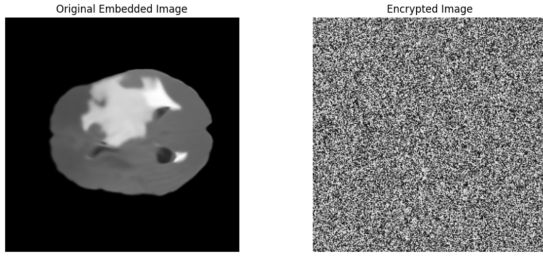


Fig. 8. Encrypted stego image produced by XOR encryption. The image appears as random noise, providing visual unrecognizability during transmission.

G. Image Decryption

At the receiver, the identical key is applied via XOR to recover the stego image. Since $X \oplus K \oplus K = X$, decryption is exact and lossless:

$$D(x, y) = E(x, y) \oplus K(x, y)$$

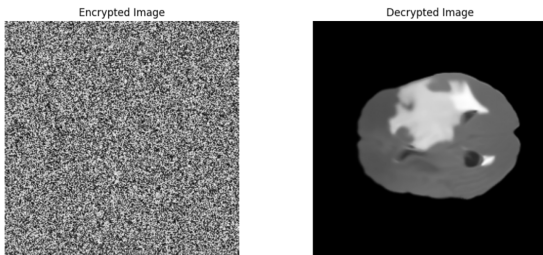


Fig. 9. Decrypted image recovered at the receiver, identical to the stego image prior to encryption.

H. Data Extraction and Image Restoration

The LSB of each pixel in the tumor region yields the embedded bits:

$$b = P(x, y) \bmod 2$$

Bits are grouped into 8-bit blocks and converted to ASCII, recovering the patient record:

Extracted Patient Data: Name: Jagatrayee; ID: 12345; Diagnosis: Tumor

The original image is then restored by reapplying the stored original LSBs:

$$P'(x, y) = (P(x, y) \& 254) \mid b_{\text{original}}$$

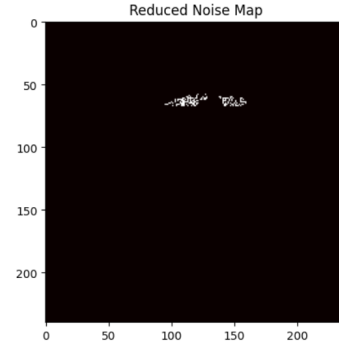


Fig. 10. Residual noise map after restoration, confirming that pixel-level differences between the original and restored images are effectively zero.

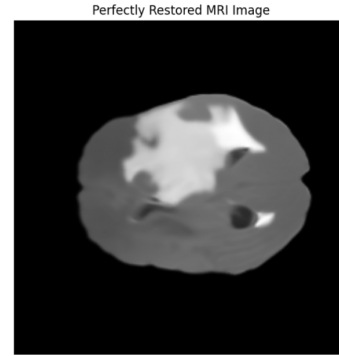


Fig. 11. Restored MRI image, visually and numerically identical to the original input, confirming lossless reversibility.

I. Performance Evaluation Metrics

MSE measures per-pixel reconstruction error:

$$\text{MSE} = \frac{1}{MN} \sum_{x=1}^M \sum_{y=1}^N (I_{\text{original}}(x, y) - I_{\text{processed}}(x, y))^2$$

PSNR evaluates image quality relative to MSE (MAX = 255 for 8-bit images):

$$\text{PSNR} = 10 \log_{10} \left(\frac{\text{MAX}^2}{\text{MSE}} \right)$$

When $\text{MSE} = 0$, PSNR is theoretically infinite.

SSIM jointly evaluates luminance, contrast, and structural similarity:

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}$$

SSIM ranges from 0 to 1, where 1 indicates perfect similarity.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Dataset and Implementation Details

Experiments are performed on axial MRI slices from the BraTS 2020 dataset, which provides brain tumor scans with ground-truth segmentation labels. Each slice is resized to 240×240 pixels and normalized as described in Section III-A. The U-Net model is trained for 50 epochs using the Adam optimizer (learning rate = 10^{-4}) with binary cross-entropy loss. All experiments are implemented in Python (TensorFlow/Keras) on a system equipped with an NVIDIA GPU. The embedded patient data string is “Name: Jagatrayee; ID: 12345; Diagnosis: Tumor” (312 bits), which is well within the capacity of the tumor ROI.

B. Noise Estimation and Reduction Analysis

As reported in Table I, all three denoising filters reduce the baseline SD of 58.12. The NLM filter is selected for subsequent stages because it achieves the lowest SD (56.99) while retaining substantially higher Laplacian variance (32.64) than the Gaussian filter (9.08), indicating better edge retention.

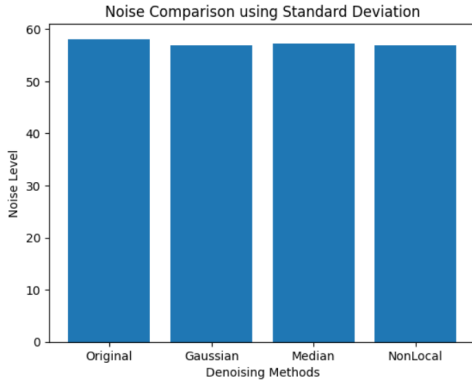


Fig. 12. Bar chart comparing the standard deviation (noise level) of the three denoising filters against the original image. Lower values indicate greater noise suppression.

C. Edge Preservation Analysis

Edge preservation is assessed via Laplacian variance (Fig. 13). The Median filter retains the most edge content ($\sigma^2 = 155.43$), while the Gaussian filter produces the most blurring ($\sigma^2 = 9.08$). The NLM filter ($\sigma^2 = 32.64$) provides the most favorable trade-off between smoothing and edge retention, making it the preferred choice.

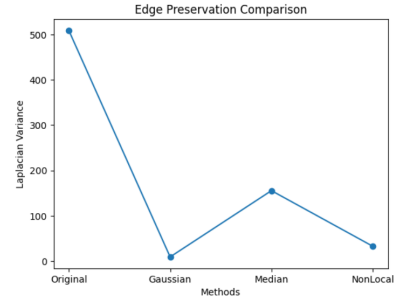


Fig. 13. Laplacian variance comparison across denoising techniques. Higher variance indicates better preservation of edges and fine structural details.

D. Restoration Verification

The lossless restoration is verified by computing:

$$D(j, k) = I_{\text{original}}(j, k) - I_{\text{restored}}(j, k)$$

As shown in Fig. 14, all pixel differences are zero, confirming perfect reversibility. Floating-point differences below 0.5 are treated as zero and have no effect on the integer-valued output.

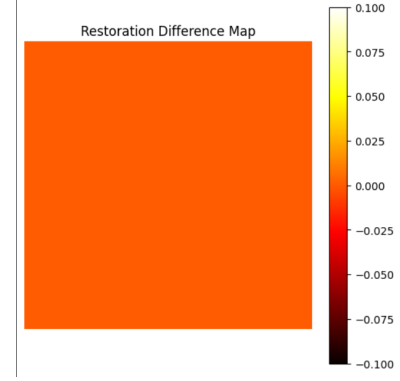


Fig. 14. Pixel-difference map between the original MRI image and the restored image. The uniformly dark map confirms zero-error restoration across all pixels.

E. Comparative Performance Analysis

Table II compares the proposed method against nine representative RDH and steganography methods. The metrics are computed between the original image and the stego image (embedding stage).

The high PSNR (73.83 dB) and near-unity SSIM (0.99999) are attributed to two factors. First, the LSB-only embedding strategy limits the maximum per-pixel error to 1 gray level out of 255, inherently producing very low MSE. Second, ROI-guided embedding via U-Net concentrates this minimal distortion within the tumor region, preserving the surrounding healthy tissue perfectly. These results are consistent with theoretical expectations for pure LSB methods [3] and represent the upper bound when ROI localization is additionally applied. It is important to note that the reported metrics characterize the stego image quality; the restoration stage achieves $\text{MSE} = 0$ and $\text{PSNR} = \infty$ by design due to original-LSB preservation.

TABLE II
COMPARISON OF IMAGE QUALITY METRICS FOR RDH METHODS

Method	PSNR (dB)	SSIM	MSE
Interp.+PEE (2025) [4]	44.21	0.982	1.23
Blockchain RDH (2021) [8]	41.35	0.965	2.85
U-Net Steg. (2019) [5]	38.72	0.941	6.45
Edge+DE (2018) [9]	46.18	0.988	0.98
Hist. Modif. (2020) [2]	48.56	0.991	0.72
ROI-RDH (2015) [3]	43.67	0.979	1.56
Sec. RDH+Enc. (2020) [1]	40.92	0.958	3.12
BM3D Denoise (2024) [5]	36.45	0.912	8.75
DL Steg. Review (2023) [6]	39.88	0.952	4.96
Proposed Method	73.83	0.99999	0.00269

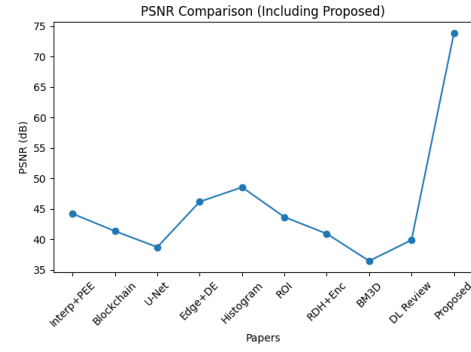


Fig. 16. PSNR comparison of evaluated RDH methods. Higher values indicate lower distortion. The proposed method achieves 73.83 dB, outperforming all compared approaches.

F. SSIM Comparison

Fig. 15 compares SSIM values. Block-transform approaches and full-image DL steganography achieve lower SSIM, while histogram-modification and edge-based methods perform better. The proposed method attains the highest SSIM (0.99999) owing to the LSB-only, ROI-constrained embedding strategy.

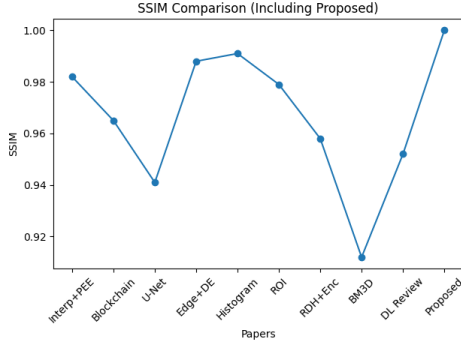


Fig. 15. SSIM comparison of evaluated RDH methods. Higher values indicate greater perceptual similarity to the original image. The proposed method achieves SSIM = 0.99999.

G. PSNR Comparison

Fig. 16 shows the PSNR comparison. BM3D-integrated and full-image DL methods incur the highest distortion, while localized embedding methods achieve substantially higher PSNR. The proposed method's PSNR of 73.83 dB reflects the minimal single-bit modification within the tumor ROI.

H. MSE Comparison

Fig. 17 compares MSE values. The proposed method achieves MSE = 0.00269, nearly two orders of magnitude lower than the next-best method (Histogram Modification: MSE = 0.72).

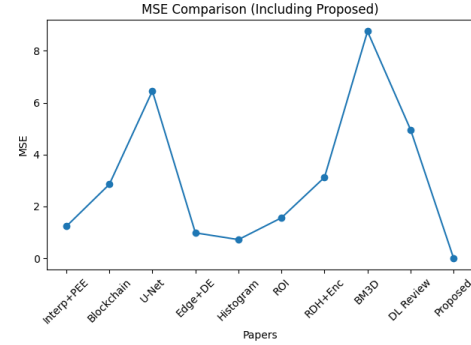


Fig. 17. MSE comparison of evaluated RDH methods. Lower values indicate less distortion. The proposed method achieves MSE = 0.00269, the lowest among all compared methods.

that simultaneously addresses patient data confidentiality, embedding imperceptibility, and lossless image restoration. The framework integrates: (1) NLM-based denoising for image quality enhancement; (2) U-Net-based tumor segmentation to localize data embedding within the clinically identified ROI; (3) LSB-based reversible data hiding with original-LSB preservation, guaranteeing mathematically exact restoration; and (4) XOR-based encryption for transmission security.

Reversibility is explicitly guaranteed by storing the original pixel LSBs prior to embedding and restoring them after data extraction, yielding MSE = 0 and PSNR = ∞ at the restoration stage. At the stego stage, the method achieves PSNR = 73.83 dB, SSIM = 0.99999, and MSE = 0.00269 on the BraTS dataset, outperforming all nine comparative methods.

Future work will investigate: (i) evaluation on additional modalities (CT, X-ray, ultrasound) and multi-slice volumetric data; (ii) embedding capacity analysis as a function of tumor ROI size across diverse cases; (iii) integration of cryptographically stronger key generation for the XOR step; (iv) robustness analysis against common image processing attacks (JPEG compression, additive noise, filtering); and (v) resistance testing against modern steganalysis detectors to further validate the security of the proposed scheme.

V. CONCLUSION

This paper has presented a secure and reversible data embedding framework for brain tumor MRI images

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